SYNOPSIS

Waste management has become one of the most critical environmental challenges in recent years due to rapid urbanization and increasing population. Improper waste disposal leads to pollution, health hazards, and environmental degradation. Traditional waste segregation methods rely heavily on manual identification, which often results in errors and inefficiency. There is a growing need for intelligent systems that can assist individuals in correctly identifying and disposing of waste. The integration of Artificial Intelligence into waste management can significantly improve accuracy and sustainability.

BINLY – Smart AI-Based Waste Classification and Management System is a web-based application developed to address this issue using modern AI technologies. The system enables users to upload images of waste materials through a simple and user-friendly interface. Once the image is uploaded, the system processes it using a trained image classification model. The application then predicts the category of waste such as recyclable, organic, or hazardous. This automated classification reduces human effort and increases segregation accuracy.

The core of the system is built using a Convolutional Neural Network (CNN) model designed for image recognition tasks. The model analyses visual patterns, textures, and shapes in the uploaded image to determine the correct waste category. The prediction is accompanied by a confidence score to indicate the reliability of the classification. This helps users understand how accurate the system's decision is. By leveraging machine learning, the application enhances decision-making in waste disposal.

The web application is developed using modern frontend technologies such as React with TypeScript and styled using Tailwind CSS to ensure responsiveness and improved user experience. The backend and database management are handled using Supabase, which provides secure data storage and authentication features. The project is deployed on a cloud platform to allow easy accessibility from any device with internet connectivity. The system architecture follows a modular design for scalability and maintainability. This ensures smooth integration between the AI model, database, and user interface.

One of the major objectives of BINLY is to create environmental awareness among users. By providing disposal suggestions along with classification results, the system educates users on proper waste management practices. The application encourages recycling and responsible waste handling habits. This can contribute to reducing landfill waste and improving

recycling efficiency. The system thus supports sustainable development goals related to environmental protection.

The proposed system overcomes several limitations of existing waste management approaches. Traditional systems lack automation, real-time feedback, and digital tracking capabilities. BINLY introduces intelligent automation, instant results, and data storage for future analysis. It reduces dependency on manual segregation and minimizes classification errors. The digital platform also opens opportunities for future enhancements such as analytics dashboards and reporting features.

BINLY demonstrates how Artificial Intelligence and web technologies can be combined to solve real-world environmental problems. The project highlights the practical implementation of machine learning in sustainability-focused applications. It provides an efficient, scalable, and user-friendly solution for smart waste management. With further enhancements such as mobile integration and multilingual support, the system can be expanded for wider adoption. This project serves as a step towards building smarter and greener communities.

CHAPTER 1

1.INTRODUCTION

Waste management has become a significant global issue due to rapid urbanization, industrial growth, and population increase. The improper disposal of waste materials leads to environmental pollution, health risks, and depletion of natural resources. Traditional waste segregation methods are often inefficient and depend heavily on manual effort. There is a strong need for intelligent and automated systems to improve waste classification accuracy. Modern technology can play a vital role in solving these environmental challenges.

Artificial Intelligence has emerged as a powerful tool in solving real-world problems through automation and smart decision-making. Image recognition technology, particularly using machine learning models, enables systems to identify objects with high accuracy. Applying AI in waste management can simplify segregation and promote responsible disposal practices. Automated classification reduces human error and saves time. This technological integration can significantly improve recycling efficiency and environmental sustainability.

BINLY – Smart AI-Based Waste Classification and Management System is developed as a web-based solution to address the issue of improper waste segregation. The application allows users to upload images of waste materials and receive instant classification results. The system predicts whether the waste is recyclable, organic, or hazardous. It also provides suggestions for appropriate disposal methods. This makes waste management more accessible and user-friendly for common people.

The system is designed using modern web technologies to ensure responsiveness and ease of use. The frontend interface is interactive and simple, allowing users to upload images effortlessly. The backend processes the image using an AI-based classification model. The prediction results are displayed along with a confidence score for transparency. The modular design ensures smooth communication between user interface, model, and database.

The main objective of this project is to promote sustainable waste management practices using technology. By providing real-time feedback and educational suggestions, the system increases environmental awareness among users. It aims to reduce landfill waste and encourage recycling habits. The project demonstrates the practical application of machine

learning in environmental solutions. Overall, BINLY contributes towards building a smarter and greener society.

1.1 ORGANIZATION PROFILE

BINLY – Smart AI-Based Waste Classification and Management System project is developed as part of an academic initiative aimed at addressing real-world environmental challenges through technology. The organization focuses on integrating Artificial Intelligence and web development to create practical and innovative solutions. It emphasizes sustainability, smart automation, and user-centered design. The primary objective is to build systems that contribute positively to society and environmental protection. This project reflects the organization's commitment to responsible technological advancement.

The development team consists of individuals with knowledge in full-stack development, database management, and machine learning concepts. The organization encourages applying theoretical learning into practical implementation through live projects. Modern tools and frameworks are used to ensure efficiency, scalability, and performance. Special focus is given to structured system architecture and modular development. This approach ensures the application is maintainable and adaptable for future upgrades.

The organization strongly promotes environmental awareness and digital transformation in waste management practices. By leveraging AI-based image classification, it aims to simplify waste segregation for common users. The project supports smart decision-making by providing real-time waste identification and disposal suggestions. It also encourages recycling habits and responsible waste handling. Through innovation, the organization contributes toward sustainable development goals.

In addition to technical development, the organization emphasizes continuous improvement and future scalability. Plans include expanding the system with advanced analytics, mobile compatibility, and multilingual support. The team values user feedback and system performance optimization. By combining environmental responsibility with emerging technologies, the organization strives to deliver impactful and socially beneficial solutions. Overall, it aims to build smarter systems for a cleaner and greener future.

1.2 SYSTEM SPECIFICATION

The BINLY – Smart AI Waste Analyzer system is designed using modern hardware and software technologies to ensure efficient performance and accessibility. The application can be accessed through any device such as a laptop, desktop, or smartphone with internet connectivity. It requires a minimum of 4GB RAM and a standard processor for smooth operation. The frontend is developed using React with TypeScript and styled using Tailwind CSS for responsive design. The backend and database management are handled using Supabase, ensuring secure data storage and authentication. The system is deployed on a cloud platform, enabling scalable performance and easy maintenance.

1.2.1 Hardware Configuration

Component	Specification
Processor	Intel Core i7 (10th Generation)
CPU Speed	Up to 4.90 GHz
System Architecture	64-bit Operating System, x64-based Processor
Main Memory (RAM)	16 GB DDR4
Storage	1 TB NVMe Solid State Drive (SSD)
Graphics	Integrated Intel UHD Graphics / Dedicated GPU (Optional)
Input Devices	Keyboard, Mouse, HD Webcam
Output Devices	LED Monitor / LCD Display

1.2.2 Software Configuration

Component	Specification
Operating System	Windows 10 / Windows 11 (64-bit)

Component	Specification
Frontend Technology	React.js with TypeScript
Styling Framework	Tailwind CSS
Backend Service	Supabase (Backend-as-a-Service)
Database	PostgreSQL
Development Tools	Visual Studio Code
Build Tool	Vite
Browser Support	Google Chrome / Microsoft Edge
Deployment Platform	Vercel Cloud Platform

1.3 FRONTEND

The frontend of the BINLY system is designed to provide a responsive, interactive, and user-friendly interface for waste image upload and result display. It ensures smooth navigation, real-time updates, and compatibility across multiple devices and browsers.

1.3.1 Frontend Technologies Used

Technology	Description
React.js	A JavaScript library used to build dynamic and component-based user interfaces.
TypeScript	A typed superset of JavaScript that improves code reliability and maintainability.
Tailwind CSS	A utility-first CSS framework used to design responsive and modern UI layouts.

Technology	Description
Vite	A fast build tool used for frontend development and optimized project bundling.
HTML5	Used to structure the web application content.
CSS3	Used for styling and enhancing the visual appearance of the application.

1.3.2 BACKEND

The backend of the BINLY system handles image processing, AI model integration, database operations, and secure communication between the frontend and server. It ensures proper request handling, data validation, and storage of classification results for efficient system functionality.

1.3.3 Backend Technologies Used

Technology	Description
Supabase	Backend-as-a-Service platform used for database management, authentication, and API handling.
PostgreSQL	Relational database used to store waste classification records and user data securely.
Node.js	JavaScript runtime used to handle server-side logic and API requests.
REST API	Enables communication between frontend and backend for data processing and retrieval.
JWT Authentication	Provides secure user authentication and protected data access.
Cloud Deployment (Vercel)	Used to host and manage backend services with scalability and reliability.

1.3.4 Frontend Features

The frontend of the BINLY web application is designed to provide a simple, interactive, and user-friendly experience. It ensures smooth navigation and responsive design across different devices such as mobile phones, tablets, and desktops.

- Responsive and modern UI design
- Image upload and preview functionality
- Real-time waste classification result display
- Confidence score visualization
- Disposal suggestion and guidance section
- Smooth navigation with structured pages
- Error handling and input validation
- Interactive buttons and user feedback messages

1.3.5 Backend Features

The backend of the BINLY system manages data processing, AI integration, and database operations. It ensures secure communication between the frontend interface and the AI model for accurate waste classification.

- AI-based image classification processing
- API integration for model communication
- Secure database storage (user uploads and results)
- Authentication and user management
- Server-side validation of uploaded files
- Data logging and record maintenance
- Cloud deployment and scalability support
- Secure data handling and access control mechanisms

CHAPTER 2

2. SYSTEM STUDY

System study is an important phase in software development that involves analysing the current system and identifying its limitations. It helps in understanding user requirements and technical feasibility before designing a new solution. This phase focuses on examining how existing waste management practices operate. Proper system study ensures that the proposed solution effectively addresses real-world problems. It forms the foundation for successful system design and implementation.

In the existing waste management system, segregation of waste is mainly performed manually by individuals or sanitation workers. This process depends heavily on human judgment and awareness. Many users are unaware of proper waste categories such as recyclable, organic, or hazardous materials. As a result, improper segregation leads to environmental pollution and reduced recycling efficiency. The lack of automation limits accuracy and scalability.

The traditional system also lacks digital monitoring and tracking mechanisms. There is no centralized platform to record or analyse waste classification data. Without data storage, it becomes difficult to evaluate waste patterns or improve recycling strategies. Manual processes consume more time and effort, especially in urban areas with high waste generation. These limitations highlight the need for a smarter and automated approach.

The proposed system, BINLY – Smart AI Waste Analyzer, is designed to overcome the drawbacks of the existing system. It introduces Artificial Intelligence to automate waste classification through image recognition. Users can upload waste images and receive instant category predictions. The system provides confidence scores and disposal suggestions to guide users. This improves accuracy and reduces manual effort.

The system study also evaluates the technical feasibility of implementing AI in a web-based environment. Modern technologies such as React, cloud databases, and machine learning models make this integration possible. The application ensures accessibility through internet-enabled devices. Scalability and modular design allow future enhancements and performance improvements. Thus, the system is technically feasible and sustainable.

Overall, the system study clearly indicates the necessity of an intelligent waste classification platform. By analyzing both existing limitations and proposed advantages, the project demonstrates practical relevance. The integration of AI enhances efficiency and promotes environmental awareness. The study confirms that the proposed system is both technically viable and socially beneficial. This forms a strong basis for further system design and development.

2.1 EXISTING SYSTEM

The existing waste management system mainly depends on manual segregation of waste materials. Individuals are responsible for identifying whether waste is recyclable, organic, or hazardous. This process relies on personal awareness and knowledge, which may not always be accurate. As a result, improper waste disposal is common.

Traditional systems do not use automated tools or intelligent classification methods. There is no AI-based mechanism to analyze waste type instantly. Users often mix different categories of waste due to lack of proper guidance. This reduces recycling efficiency and increases landfill waste.

In most areas, waste segregation awareness programs are limited. People may not clearly understand disposal rules or color-coded bin systems. There is also no digital platform to assist users in real-time decision-making. The absence of technological support makes the system less effective.

Overall, the existing system lacks automation, accuracy, and user assistance. It is time-consuming and prone to human error. There is no proper tracking or data storage for waste classification patterns. These limitations highlight the need for an intelligent AI-based waste analysis system like BINLY.

2.1.1 DRAWBACKS OF THE EXISTING SYSTEM

The existing waste management system suffers from heavy dependence on manual segregation. Since classification is done by individuals, the accuracy depends on their knowledge and awareness. Many people are unaware of correct waste categories. This results in improper disposal and contamination of recyclable materials.

Another major drawback is the absence of automation and real-time assistance. There is no intelligent system to guide users instantly while disposing of waste. Without technological

support, mistakes in segregation are common. This reduces overall recycling efficiency and increases environmental pollution.

The traditional system also lacks digital record maintenance and monitoring. There is no centralized platform to track waste classification data or analyze patterns. Authorities cannot easily measure waste generation trends. This limits strategic planning for effective waste management.

Additionally, improper segregation leads to increased landfill waste and resource wastage. Hazardous waste may get mixed with general waste, creating health risks. The system is time-consuming and inefficient in large urban areas. These drawbacks emphasize the need for an AI-based smart waste analysis system like BINLY.

2.2 PROPOSED SYSTEM

The proposed system, BINLY – Smart AI-Based Waste Classification and Management System is a web-based application that automates waste classification using Artificial Intelligence. Users can upload images of waste materials through a simple and responsive interface. The system processes the image using a trained machine learning model. It then predicts the appropriate waste category such as recyclable, organic, or hazardous.

The application integrates frontend, backend, and AI modules in a structured architecture. The frontend ensures smooth user interaction, while the backend handles image processing and database operations. The AI model analyzes visual patterns to generate accurate predictions with a confidence score. This reduces manual effort and improves segregation accuracy.

The system also provides proper disposal suggestions based on the predicted category. It promotes environmental awareness by guiding users toward responsible waste management practices. The platform is cloud-deployed, ensuring accessibility and scalability. Overall, the proposed system offers a smart and efficient solution for sustainable waste management.

2.2.1 FEATURES OF THE PROPOSED SYSTEM

The proposed BINLY system introduces AI-based automatic waste classification using image recognition technology. Users can upload an image of waste material, and the system instantly predicts its category. The classification is performed using a trained machine learning model. This reduces human effort and improves segregation accuracy.

The system provides real-time results along with a confidence score. This helps users understand the reliability of the prediction. In addition to classification, the application offers proper disposal suggestions. These suggestions promote eco-friendly waste management practices.

Another key feature is a responsive and user-friendly web interface. The application can be accessed from mobile phones, tablets, and desktops. Input validation and error handling ensure smooth user interaction. The system is designed with a clean and simple UI for easy navigation.

The backend securely stores uploaded images and classification results in a database. This allows data tracking and future analysis of waste patterns. The system is cloud-deployed, ensuring scalability and accessibility. Overall, the proposed system enhances efficiency, awareness, and sustainability in waste management.

2.2.2 ADVANTAGES OF PROPOSED SYSTEM

The proposed BINLY system offers automated waste classification using Artificial Intelligence, which significantly reduces human effort and error. By analyzing images instantly, the system provides quick and accurate results. This improves segregation efficiency compared to traditional manual methods. Automation ensures consistent and reliable waste identification.

Another major advantage is real-time user guidance and disposal suggestions. The system not only classifies waste but also educates users on proper disposal practices. This increases environmental awareness and promotes responsible behavior. It supports eco-friendly waste management at the individual level.

The system is accessible through a web-based platform, making it easy to use from any internet-enabled device. Its responsive design ensures smooth performance on mobile, tablet, and desktop devices. Cloud deployment provides scalability and reliable access. This makes the system practical for wider adoption.

Additionally, the backend stores classification data for tracking and future analysis. Authorities or developers can analyze waste patterns and improve strategies. The modular architecture allows easy upgrades and feature expansion. Overall, the proposed system enhances efficiency, sustainability, and technological innovation in waste management.

2.3 FEASIBILITY STUDY

Feasibility study is conducted to evaluate whether the proposed BINLY system is practical and implementable. It analyzes technical, economic, and operational aspects of the project. This study ensures that the system can be successfully developed and deployed without major risks.

2.3.1 TECHNICAL FEASIBILITY

The BINLY system is technically feasible because it uses modern and widely supported technologies such as React, Supabase, and cloud deployment platforms. Image classification using machine learning models like CNN is already proven in real-world applications. The required hardware and software resources are easily available. Integration between frontend, backend, and AI modules is achievable with existing tools and APIs. Therefore, the technical implementation of the system is practical and achievable.

2.3.2 ECONOMIC FEASIBILITY

The project is economically feasible as it uses open-source technologies and cloud-based services. Development tools like React and Supabase reduce infrastructure costs. Deployment on platforms such as Vercel minimizes hosting expenses. Since it is a web-based system, no expensive hardware installation is required for users.

2.3.3 OPERATIONAL FEASIBILITY

The BINLY system is operationally feasible because it is user-friendly and easy to access. Users only need an internet-enabled device to operate the application. The interface is simple, requiring minimal technical knowledge. This ensures smooth adoption and effective utilization of the system.

2.4 SYSTEM WORKING OVERVIEW

The BINLY – Smart AI-Based Waste Classification and Management System is designed to simplify the process of identifying waste materials using artificial intelligence. The system allows users to upload images of waste items through a web interface. Once the image is uploaded, the application processes it using AI-based image classification techniques.

The backend system analyzes the visual features of the image such as shape, texture, and patterns. Based on this analysis, the system predicts the category of waste such as recyclable, organic, or hazardous. The result is then displayed to the user along with disposal recommendations.

The working overview of the system helps users understand how the BINLY platform processes data and provides intelligent waste management guidance. This improves user awareness and encourages responsible waste disposal practices.

- The user accesses the BINLY web application through a browser.
- The user logs into the system using email and password authentication.
- The user selects the preferred language for interacting with the application.
- The user uploads an image of the waste item or provides voice input.
- The system validates the input data and sends it to the backend server.
- The AI model analyzes the waste image and identifies the waste category.
- The system generates classification results along with disposal suggestions.
- The result is displayed to the user in the selected language.

CHAPTER - III

3. SYSTEM DESIGN AND DEVELOPMENT

System design and development is a crucial phase in the BINLY project where the conceptual idea is transformed into a functional web application. This phase defines the overall architecture, modules, and data flow of the system. A structured design approach ensures smooth integration between frontend, backend, and AI components. Proper planning improves scalability, maintainability, and system performance.

The system architecture follows a modular design pattern separating user interface, server logic, and database management. The frontend is developed using modern web technologies to provide a responsive and interactive user experience. The backend handles image processing, API communication, and secure data storage. The AI model is integrated to analyze uploaded images and generate accurate waste classification results.

During development, emphasis was placed on usability, performance optimization, and secure communication. Input validation mechanisms are implemented to ensure correct image formats and prevent errors. The backend ensures secure handling of user data and proper database connectivity. Cloud deployment allows the system to be accessed from any internet-enabled device.

The development process involved coding, testing, debugging, and deployment stages. Each module was tested individually before full system integration. Continuous improvements were made to enhance accuracy and responsiveness. Overall, the system design and development phase successfully transformed the proposed concept into a practical AI-based waste analysis solution.

3.1 SYSTEM ARCHITECTURE DESIGN

The system architecture of BINLY is designed using a modular and layered approach to ensure efficient communication between different components. The architecture separates the application into frontend, backend, AI processing, and database layers. This separation improves system maintainability and scalability. Each layer performs a specific function while interacting seamlessly with others. The structured design enhances overall performance and reliability.

The frontend layer is responsible for user interaction and interface design. It allows users to upload waste images and view classification results. The interface is responsive and accessible across multiple devices. Proper input validation is implemented to ensure correct file formats. This layer communicates with the backend through secure API requests.

The backend layer manages business logic, image processing, and communication with the AI model. When a user uploads an image, the backend processes the request and sends it to the classification module. It receives the prediction results and formats them for display. The backend also ensures secure handling of user data. This layer acts as the bridge between frontend and database.

The AI processing layer contains the trained machine learning model used for image classification. It analyzes visual patterns, textures, and shapes in the uploaded waste image. The model predicts the appropriate waste category with a confidence score. This intelligent processing improves classification accuracy. The AI module is integrated with the backend for seamless operation.

The database layer stores uploaded image details, predicted categories, and timestamps. It ensures data persistence for tracking and future analysis. Secure authentication and data validation mechanisms protect stored information. Cloud deployment allows centralized access and scalability. Overall, the architecture ensures smooth data flow, security, and efficient system performance.

3.2 FILE DESIGN

File design in the BINLY system ensures proper organization of project files for easy development and maintenance. The project follows a modular folder structure separating frontend, backend, and configuration files. This improves readability and simplifies debugging.

The frontend files include components, pages, and styling resources required for user interaction. Each UI element such as image upload and result display is maintained in separate component files. This promotes code reusability and clean structure.

Backend files manage API handling, AI model integration, and database connectivity. Service files are used to process image classification requests and return results. Environment files securely store configuration details and access keys.

Configuration and dependency files are maintained in the root directory for build and deployment management. These files control project compilation and cloud deployment settings. Overall, structured file design enhances scalability and system performance.

3.3 INPUT DESIGN

Input design in the BINLY system focuses on collecting user data accurately and efficiently. The primary input is the waste image uploaded by the user. The interface is designed to be simple and user-friendly.

Users can upload images in supported formats such as JPG or PNG. File validation is implemented to ensure correct format and size restrictions. This prevents system errors and ensures smooth processing.

The system also provides an image preview option before submission. This allows users to confirm the selected file. Clear instructions are displayed to guide users during the upload process.

Proper input validation is performed at both frontend and backend levels. This ensures data integrity and security. Overall, the input design enhances usability and reduces processing errors.

3.4 OUTPUT DESIGN

The output design of the BINLY system focuses on presenting classification results in a clear and understandable format. Once the user uploads an image, the system processes it and displays the predicted waste category. The result is shown prominently on the screen for easy visibility. A confidence score is also displayed to indicate the accuracy of the prediction. This ensures transparency in the system's decision-making process.

In addition to the waste category, the system provides proper disposal suggestions. These suggestions guide users on whether to recycle, compost, or dispose of the item safely. The output interface is designed using a clean and responsive layout. Visual elements such as icons or highlighted text improve readability. Error messages are displayed clearly if the input is invalid or processing fails.

The output is generated dynamically after successful AI processing. It updates instantly without requiring the user to refresh the page. The results can also be stored in the database for

record-keeping and analysis. The design ensures that information is presented in a structured and organized manner. Overall, the output design enhances user experience and promotes informed waste management decisions.

3.5 DATABASE DESIGN

The database design of the BINLY system is structured to store and manage waste classification data efficiently. It maintains records of uploaded images, predicted categories, confidence scores, and timestamps. The database ensures data consistency and integrity through proper table structure. A relational database model is used for organized data storage. This design supports easy retrieval and analysis of stored information.

The primary table includes fields such as unique ID, image URL or path, waste category, confidence percentage, and created date. Each record represents a single classification request made by a user. Proper data types are assigned to each field to maintain accuracy. Timestamp fields help in tracking when the data was generated. This structured approach ensures systematic record maintenance.

The database is securely connected to the backend using authenticated access methods. Validation mechanisms are implemented to prevent unauthorized data manipulation. Cloud-based database services provide scalability and reliability. The design also allows future expansion such as adding user accounts or analytics features. Overall, the database design supports efficient and secure system operation.

3.6 SYSTEM DEVELOPMENT

The system development phase of BINLY involves transforming the proposed design into a functional web application. The development process started with frontend interface creation using modern web technologies. A responsive and user-friendly design was implemented for smooth interaction.

The backend development focused on handling image uploads and integrating the AI classification model. APIs were created to manage communication between frontend and backend. Secure database connectivity was also established during this stage.

The AI model was integrated to analyze uploaded waste images and generate predictions. Proper validation and error-handling mechanisms were implemented to ensure reliability. Each module was tested individually before full system integration.

Finally, the application was deployed on a cloud platform for public access. Performance optimization and debugging were carried out to improve efficiency. Continuous testing ensured smooth functionality across different devices and browsers.

3.7 DESCRIPTION OF MODULES

The BINLY system consists of multiple functional modules that work together to ensure smooth operation. The User Authentication module manages secure login and access control, while the Image Processing module prepares images for analysis. The AI Classification module predicts waste categories, and the Recommendation module provides proper disposal guidance. Finally, the Database and Deployment modules ensure secure data storage, cloud hosting, and overall system reliability.

1. User Authentication Module

This module manages secure user registration and login functionality. It ensures that only authorized users can access the system features. Supabase authentication is used for session management and data protection.

For example : a user logs in before uploading waste images.

2. Image Processing Module

This module handles image validation and preprocessing before AI analysis. It checks file format, resizes images, and prepares them for classification. Proper preprocessing improves prediction accuracy.

For example : an uploaded battery image is formatted before analysis.

3. AI Classification Module

This is the core module of BINLY that performs waste classification using CNN-based concepts. It analyzes image patterns and predicts the waste category with confidence score. The module determines whether the waste is recyclable, organic, or hazardous.

For example : a plastic bottle is classified as recyclable waste.

4. Recommendation Module

This module provides disposal guidance based on the predicted waste category. It helps users understand how to properly dispose of or recycle the item. The system displays eco-friendly suggestions for better waste management.

For example: recyclable waste shows instructions to place it in a recycling bin.

5. Database Management Module

This module stores classified results, user activity, and system records in PostgreSQL via Supabase. It maintains structured data for tracking and analytics. This ensures data consistency and secure storage.

For example: previous classification history is saved for future reference.

6. Deployment & Cloud Integration Module

This module ensures smooth deployment of the application on Vercel cloud platform. It manages frontend hosting, backend connectivity, and API integration. The module guarantees scalability and high availability.

For example: users can access BINLY online without installing any software.

3.8 SYSTEM WORKFLOW

The BINLY Smart AI-Based Waste Classification and Management System follows a clear workflow to analyze waste images and offer disposal recommendations. The process begins when the user opens the web application and logs in. After logging in, the user can choose their preferred language for interaction.

Once the language is selected, the user uploads an image of a waste item or uses voice input to describe it. The uploaded data goes to the backend server, where the system checks the input and gets it ready for processing. The backend then sends the image to the Artificial Intelligence classification module.

The AI model analyzes the image with image recognition techniques to identify the type of waste, such as recyclable, organic, or hazardous. The system generates a confidence score to show how accurate the prediction is. Based on the identified category, it also offers proper disposal suggestions and eco-friendly recommendations.

Finally, the result appears for the user in the chosen language, along with instructions on how to dispose of the waste safely. The classification result can also be saved in the database for record-keeping and future analysis. This workflow ensures that the BINLY system provides quick and user-friendly waste classification.

System Workflow :

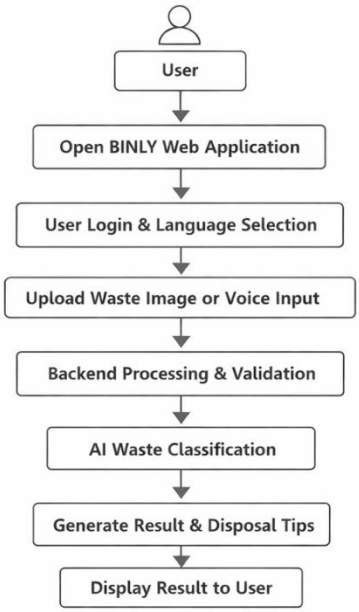

Fig: Working Flow Diagram

3.9 TECHNOLOGY SUMMARY

The BINLY Smart AI-Based Waste Classification and Management System uses modern web technologies and cloud services. The system combines frontend technologies for user interaction, backend services for data processing, and cloud platforms to host the application. These technologies ensure the system operates efficiently and remains accessible on multiple devices.

The frontend interface is built with React.js and TypeScript, which help create dynamic and interactive web pages. React lets developers build reusable components and manage the

application's structure effectively. Tailwind CSS designs responsive layouts and enhances the visual appeal of the user interface.

Backend services use Supabase for authentication, database management, and API communication. The system saves user data and classification results in a PostgreSQL database. Developers use Vite as the build tool, and they deploy the application on the Vercel cloud platform for fast performance and easy access.

Layer	Technology Used	Purpose
Frontend	React.js, TypeScript	Build dynamic user interface
Styling	Tailwind CSS	Responsive UI design
Backend	Supabase	Handle authentication and APIs
Database	PostgreSQL	Store waste classification data
Build Tool	Vite	Fast development environment
Deployment	Vercel	Cloud hosting platform

3.10 SYSTEM WORKING STEPS

The working process of the BINLY system involves a series of operations that enable the user to easily classify the waste materials. The working process is as follows:

- **User Access:** The user can access the BINLY application using the browser on the mobile device, tablet, laptop, or computer.
- **User Authentication:** The user can log in to the application using the email and password provided. The user can also register if they do not have an account.
- **Language Selection:** The user can choose the language they want to use to communicate with the application.
- **Image Upload or Voice Input:** The user can upload the image of the waste material and can also input the voice commands to describe the waste material.
- **Data Validation:** The application validates the data and checks whether it is in the correct format.

- **AI-Based Waste Analysis:** The application uses the image classification module to analyze the image using the AI algorithm.
- **Waste Classification Result:** The application identifies the type of waste and can classify it as recyclable, organic, and hazardous.
- **Recommendation Generation:** Based on the result obtained from the image classification, the application generates the recommendations.
- **Result Display:** The final result is displayed to the user.
- **Data Storage:** The results can be stored in the database for future analysis and monitoring.

3.11 LIMITATIONS AND PERFORMANCE ANALYSIS

The BINLY system offers an effective way to classify waste using artificial intelligence and image recognition technology. Nevertheless, certain limitations exist in the current BINLY system implementation. These limitations are mainly dependent on the images fed to the system and the dataset used in the development of the AI model.

One limitation is linked to the clarity of images fed to the system. If the uploaded image is not clear or contains multiple images of waste, the AI model may not classify the waste object well. This is mainly dependent on the quality of input images fed to the system by the user.

Another limitation is linked to the dataset used in the development of the AI model. If the dataset does not include certain types of waste, the AI model may not classify them well. This can be addressed by increasing the dataset used in the development of the AI model, thereby increasing the accuracy of the system.

Despite these limitations, the BINLY system works efficiently for performing basic waste classification tasks. The BINLY system works efficiently by processing images uploaded to it in a very short time. The user interface of BINLY works efficiently on different devices such as laptops, desktop computers, and mobile phones

Performance Metrics :

Performance Factor	Description
Prediction Accuracy	AI model provides classification with confidence score
Processing Speed	Results generated within seconds
Accessibility	Works on mobile, tablet, and desktop devices
Scalability	Cloud hosting supports multiple users
Reliability	Secure backend ensures stable operation

CHAPTER - IV

4. SOFTWARE TESTING AND IMPLEMENTATION

Software testing is an essential phase in the development of the BINLY system to ensure accuracy, reliability, and performance. Various testing methods such as unit testing, integration testing, and system testing were conducted to verify each module. The AI classification results were validated using sample waste images to check prediction accuracy. Testing ensured that the system functions correctly without errors or performance issues.

During implementation, the frontend was developed using React with TypeScript and styled using Tailwind CSS. The backend services and database management were handled through Supabase with PostgreSQL integration. The application was built and optimized using Vite for faster performance. Each module was integrated step by step to ensure smooth communication between frontend and backend.

After successful testing and integration, the system was deployed on the Vercel cloud platform. Deployment testing was performed to verify browser compatibility and responsiveness across devices. Security features such as authentication and secure API communication were validated. The final implementation ensures that BINLY operates efficiently in a real-time cloud environment.

4.1 TESTING METHODS

Testing methods are used to verify that the BINLY system functions correctly and meets user requirements. Different levels of testing were performed to identify and eliminate errors during development. These methods ensure reliability, accuracy, and smooth performance of the application.

Various testing techniques such as Unit Testing, Integration Testing, and System Testing were applied. Each testing stage focused on different components of the system. This structured approach improved overall software quality.

Testing also helped in validating AI prediction accuracy and database connectivity. It ensured proper frontend and backend communication. As a result, the system performs efficiently in real-time usage.

Proper documentation of test cases was maintained throughout development. Errors identified during testing were corrected immediately. This reduced the risk of failure after deployment.

4.1.1 UNIT TESTING

Unit Testing was performed to test individual modules of the BINLY system independently. Each module such as authentication, image processing, and database storage was verified separately. This ensured that every component functions as expected without errors. For example, the login module was tested to validate correct and incorrect user credentials.

4.1.2 INTEGRATION TESTING

Integration Testing was conducted to verify communication between different modules. It ensured that the frontend properly interacts with backend services and the database. Data flow between image upload, AI prediction, and result display was carefully tested. For example, after uploading an image, the prediction result was checked for correct display.

4.1.3 SYSTEM TESTING

System Testing was performed to evaluate the complete application as a whole. The entire BINLY system was tested in a real-time environment. Performance, security, and user interface responsiveness were verified. For example, multiple users accessing the system simultaneously were tested for stability.

4.2 SYSTEM IMPLEMENTATION

The system implementation phase involved developing the frontend using React with TypeScript and styling with Tailwind CSS. Backend services and database management were implemented using Supabase and PostgreSQL. Each module was developed and integrated step by step.

After development, the application was tested locally to ensure proper functionality. Errors and bugs identified during testing were resolved. The build tool Vite was used to optimize performance.

Finally, the system was deployed on the Vercel cloud platform. Deployment testing ensured browser compatibility and responsiveness. The application is now accessible online for users.

4.3 SYSTEM MAINTENANCE

System maintenance ensures continuous performance and reliability of the BINLY application. Regular monitoring is performed to check for system errors and performance issues. Necessary updates are applied when required.

Database maintenance includes managing stored records and ensuring data security. Backup procedures are implemented to prevent data loss. This improves system stability and reliability.

Future maintenance may include updating AI models and adding new features. Security updates will be applied to protect user data. Continuous improvement ensures long-term usability of the system.

4.4 TEST CASES

Testing plays an important role in ensuring that the BINLY system functions correctly. Test cases are used to verify whether different components of the application perform as expected. Each feature of the system is tested using specific input conditions and expected outputs.

Table

Test Case	Input	Expected Result
Login Test	Valid email and password	User successfully logs into the system
Image Upload Test	Upload waste image	Image accepted and processed
AI Classification Test	Waste image input	Correct waste category predicted
Result Display Test	Processed waste image	Classification result displayed

4.5 USER ACCEPTANCE TESTING

User Acceptance Testing (UAT) is conducted to verify whether the BINLY system meets user expectations and functional requirements. In this stage, real users interact with the application and evaluate its usability and performance.

Users test various features such as login functionality, image upload, AI waste classification, and result display. Feedback collected during this phase helps identify potential improvements in system usability and functionality.

User Acceptance Testing ensures that the system is reliable, easy to use, and suitable for real-world implementation.

CHAPTER – V

DATA VISUALIZATION AND ANALYSIS

The BINLY Waste Analyzer system analyzes waste images and voice inputs to automatically identify the type of waste using machine learning techniques. After the analysis process, the system generates results such as waste category, prediction confidence score, and usage records. These results are stored in the system database for further reference and evaluation. By collecting and storing this information, the application can provide useful insights about the waste classification process. The stored data can then be used for visual representation and system analysis. Visualization helps users easily understand the output generated by the system. It also improves the clarity of the results presented in the application.

Data visualization plays an important role in presenting the classification results of the BINLY Waste Analyzer in an easy and understandable format. Charts and graphs such as bar charts, pie charts, and line charts can be used to represent the distribution of waste categories, system usage trends, and prediction confidence levels. These visual representations allow users and developers to quickly analyze the performance of the system. Visualization also helps in identifying patterns and trends in waste classification data. It improves decision-making related to waste management and system improvements. By presenting results visually, the system becomes more informative and user-friendly. Therefore, visualization is an important component in analyzing and presenting the results of the BINLY Waste Analyzer system.

5.1 Waste Category Count

Bar Chart :

This bar chart represents the number of waste items classified into different categories by the BINLY Waste Analyzer system. The system analyzes waste images and voice inputs to identify the type of waste using machine learning techniques. After the analysis process, each detected waste item is categorized into groups such as recyclable, organic, and hazardous waste. The bar chart clearly shows the count of items detected in each category. It helps in comparing which waste type appears more frequently in the system. This visualization makes it easier to understand the distribution of waste types detected by the application. It also helps

in identifying patterns in waste classification results. The chart provides a simple and clear comparison between different waste categories. By analyzing this chart, users can quickly understand the waste composition handled by the system. This visualization improves the clarity and presentation of system output.

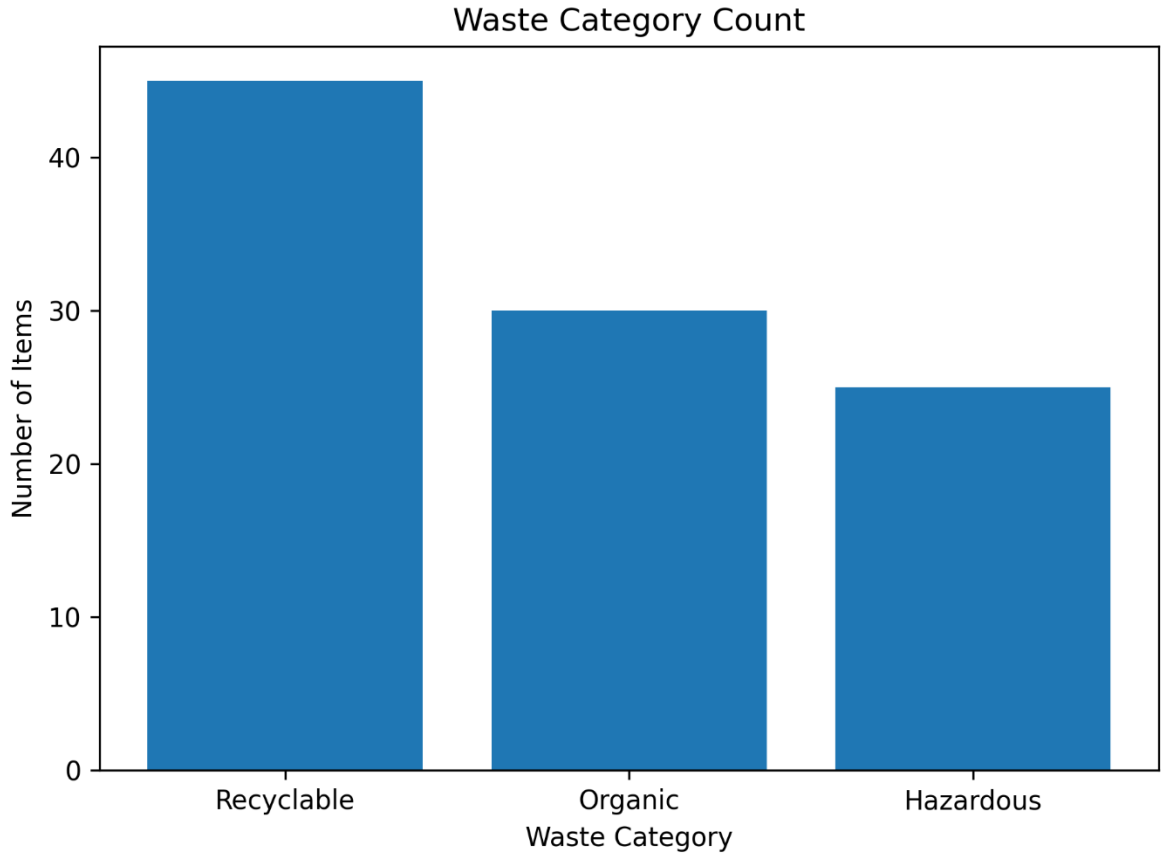

Fig: Waste Category Count

5.2 Waste Category Percentage

Pie Chart:

the BINLY Waste Analyzer system. After analyzing waste images, the system categorizes them into recyclable, organic, and hazardous waste types. The pie chart visually displays how much each category contributes to the total waste detected by the application. Each section of the pie represents the proportion of a specific waste type in percentage form. This type of visualization makes it easier to understand the overall waste distribution at a glance. The 3D representation of the chart enhances the clarity and visual appeal of the data. It allows users to quickly identify which category occupies the largest portion of the total waste.

Such visualization helps in analyzing waste management patterns and improving awareness about proper waste segregation. The chart also provides a quick overview of how different waste categories are distributed in the analyzed dataset. Overall, the pie chart improves the presentation and understanding of waste classification results.

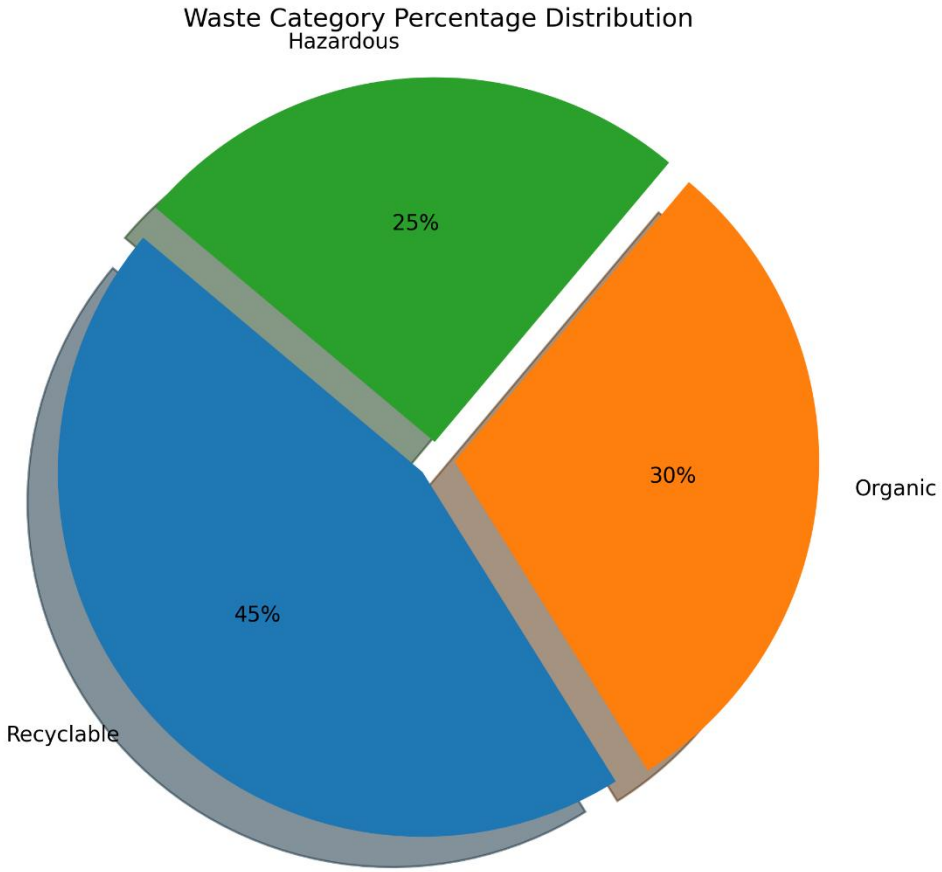

Fig: Waste Classification Category

5.3 Waste Upload Trend

Line Chart :

This line chart illustrates the trend of waste image uploads to the BINLY Waste Analyzer system over a specific period of time. The system allows users to upload waste images or capture them using a camera for classification. Each upload is recorded by the system and stored for analysis. The line chart represents how the number of uploads changes across different days. This helps in understanding the usage pattern and engagement level of users

with the application. By observing the trend line, it becomes easier to identify whether system usage is increasing or decreasing over time. It also helps developers evaluate how actively the application is being used by users. Visualization of upload trends provides valuable insights into the performance and adoption of the system. This chart also helps in monitoring system activity and detecting usage patterns. Overall, the line chart provides a clear view of the growth and interaction with the BINLY Waste Analyzer system.

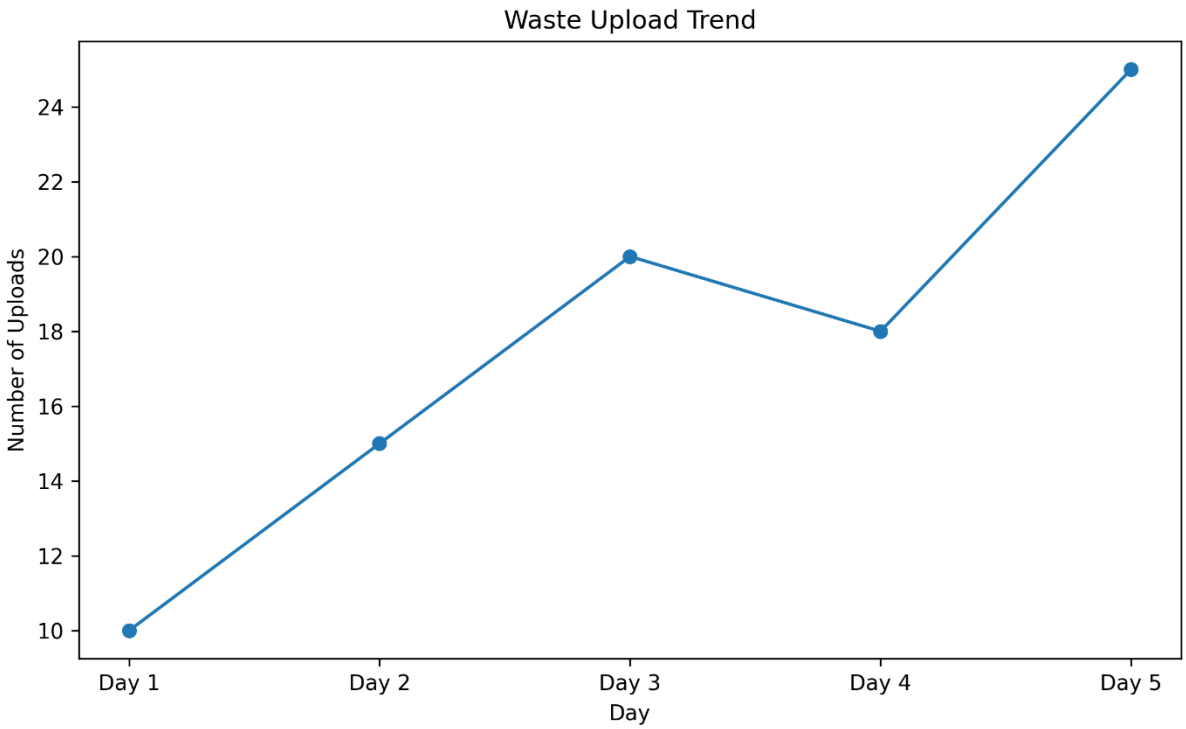

Fig: Waste Upload Trend

5.4 Prediction Confidence Score

Bar Chart :

This bar chart displays the average confidence score of the machine learning model used in the BINLY Waste Analyzer system. The confidence score indicates how certain the model is about its prediction for a particular waste category. After analyzing a waste image, the system predicts the waste type and assigns a confidence value to the prediction. The chart shows the average confidence score for each waste category such as recyclable, organic, and hazardous waste. Higher confidence scores indicate that the model is more certain about its classification results. This visualization helps in evaluating the performance and reliability of

the machine learning model used in the system. It also allows developers to compare prediction accuracy across different waste categories. By analyzing this chart, improvements can be made to enhance the classification model. The chart therefore plays an important role in understanding the prediction capability of the system. Overall, it helps in assessing the effectiveness and reliability of the waste classification process.

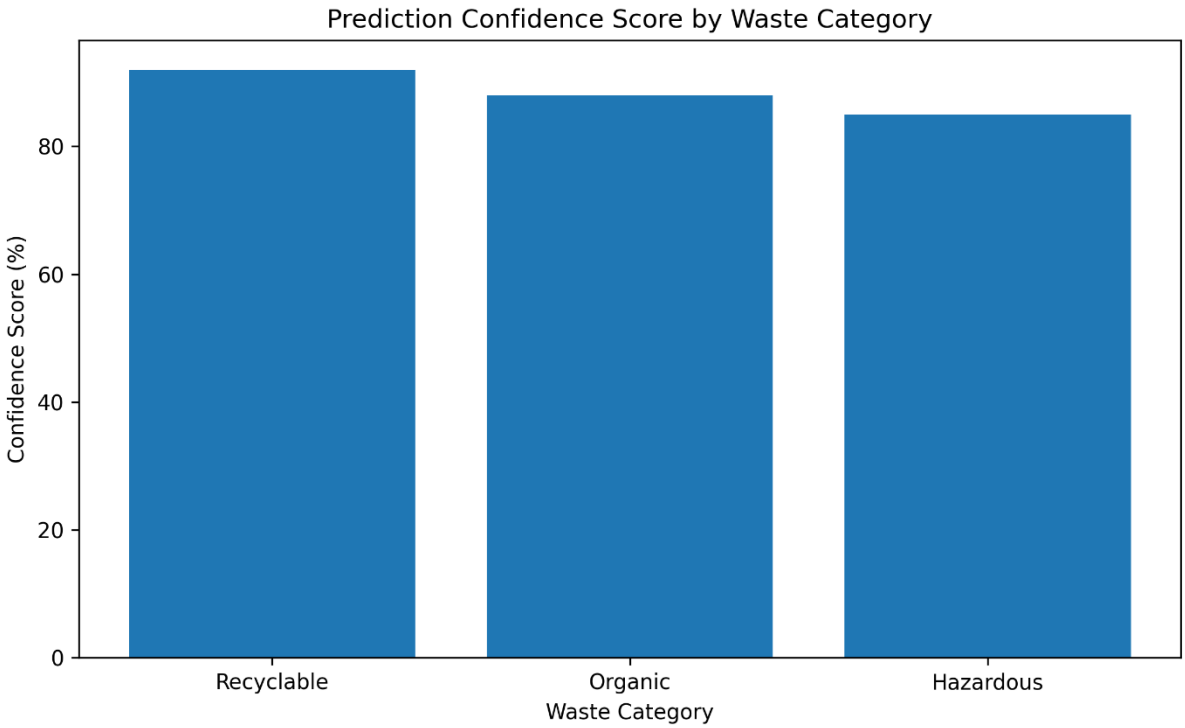

Fig: Confidence Score

CHAPTER – VI

CONCLUSION AND FUTURE ENHANCEMENT

CONCLUSION

The BINLY system successfully demonstrates the implementation of an AI-powered waste classification web application aimed at promoting smart waste management practices. The project integrates modern web technologies with machine learning concepts to automate the process of waste identification and segregation. By allowing users to upload waste images and receive instant classification results, the system improves awareness about proper disposal methods. The use of CNN-based image classification concepts enhances prediction accuracy and system intelligence. The frontend developed using React with TypeScript ensures a responsive and user-friendly interface. Backend services managed through Supabase provide secure authentication and reliable database storage. Cloud deployment on Vercel ensures scalability and real-time accessibility. Various testing methods were conducted to guarantee system stability and performance. The modular architecture design improves maintainability and future scalability. The project addresses real-world environmental challenges through a practical technological solution. It also demonstrates the effective integration of AI within a full-stack web application. Although the current version focuses on single image classification, it can be further enhanced with advanced object detection techniques. Future improvements may include real-time camera integration and mobile application support. Overall, BINLY serves as a socially impactful, technically sound, and scalable AI-based solution for sustainable waste management.

FUTURE ENHANCEMENT

The BINLY system can be further enhanced by integrating advanced AI models such as YOLO-based real-time object detection for identifying multiple waste items in a single image. Future versions can include live camera support to enable instant waste classification without manual image upload. The prediction accuracy can be improved by training the model with a larger and more diverse waste image dataset. A mobile application version can be developed to increase accessibility and user engagement. Real-time analytics dashboards can be added to monitor waste trends and user activity. Integration with IoT-enabled smart bins can automate waste segregation at the hardware level. Multi-language support can be implemented

to make the system accessible to a wider audience. A voice assistant feature can be introduced to read out disposal instructions for better usability. Advanced reporting features can help organizations track recycling performance. Cloud scalability improvements can support high user traffic efficiently. Role-based admin panels can be added for better system management. Security enhancements such as advanced encryption and monitoring can further protect user data. API integration with municipal waste management systems can improve practical implementation. Continuous model retraining can ensure up-to-date prediction performance. Overall, these enhancements will transform BINLY into a more intelligent, scalable, and industry-ready smart waste management platform.

7. REFERENCES

1. Deep Learning – Ian Goodfellow, Yoshua Bengio, Aaron Courville, MIT Press, 2016.
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7. Pattern Recognition and Machine Learning – Christopher M. Bishop, Springer, 2006.
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9. Machine Learning: A Probabilistic Perspective – Kevin P. Murphy, MIT Press, 2012.
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APPENDICES

The appendices section includes supplementary materials that support the development and implementation of the BINLY system. It contains additional documents such as sample input images, test case tables, system screenshots, and database schema details. Technical configurations, source code snippets, and deployment settings are also provided for reference. These materials help in better understanding the project's practical execution and validation process.

APPENDICES A – DATA FLOW DIAGRAM (DFD)

The Data Flow Diagram (DFD) illustrates the flow of data within the BINLY system from input to output. It shows how users upload waste images through the frontend interface. The image is then sent to the backend processing layer for validation and AI-based classification. The AI module analyzes the image and generates the predicted waste category. The result is forwarded to the database for storage and record maintenance. Simultaneously, the prediction output is sent back to the frontend for display. The DFD clearly represents the interaction between user, system processes, and database. This diagram helps in understanding the logical data movement and system structure.

APPENDICES B – DATABASE / TABLE STRUCTURE

The database structure of the BINLY system is designed to store user information, image records, and classification results in a structured manner. The Users table contains fields such as user_id, name, email, and authentication details. The Waste_Records table stores image_url, predicted_category, confidence_score, and timestamp for each uploaded image. Foreign key relationships are maintained to link waste records with corresponding users. This structured table design ensures data integrity, secure storage, and efficient retrieval of classification history.

APPENDICES C – SAMPLE CODING

This appendix includes sample source code snippets used in the development of the BINLY system. It contains frontend code for image upload functionality and result display using React with TypeScript. Backend integration code for connecting to Supabase and storing prediction results is also provided. Basic AI prediction logic and API handling are demonstrated

for reference. These code samples help in understanding the practical implementation of the system modules.

APPENDICES D – SAMPLE INPUT SCREENS

This appendix presents sample input screens of the BINLY web application. It includes screenshots of the login page, image upload interface, and user dashboard. The input screen allows users to select and upload waste images for classification. These screens demonstrate the user-friendly design and responsive layout of the system. The provided samples help in understanding how users interact with the application during real-time usage.

APPENDICES E – SAMPLE OUTPUT SCREENS

This appendix includes sample output screens generated by the BINLY system after image classification. It displays the predicted waste category along with the confidence score and disposal recommendation. The output screen clearly shows whether the waste is recyclable, organic, or hazardous. Visual indicators and structured result sections improve user understanding. These sample outputs demonstrate the accuracy and effectiveness of the AI-based prediction module.

APPENDICES F – SYSTEM ARCHITECTURE DIAGRAM

This appendix presents the overall system architecture diagram of the BINLY application. The diagram illustrates the interaction between the frontend interface, backend services, AI classification module, and database layer. It shows how user inputs flow through the processing layers and generate prediction results. Cloud deployment and authentication components are also represented in the architecture. The diagram provides a clear visual understanding of the system's structural design and workflow.

APPENDICES G – PROCESS FLOW DIAGRAM

This appendix illustrates the step-by-step process flow of the BINLY system from user interaction to result generation. It begins with user authentication, followed by image upload and validation. The uploaded image is then processed and sent to the AI classification module for prediction. The predicted result is stored in the database and displayed to the user with disposal recommendations. The process flow diagram clearly represents the sequential operations and logical workflow of the entire system.

A. DATA FLOW DIAGRAM

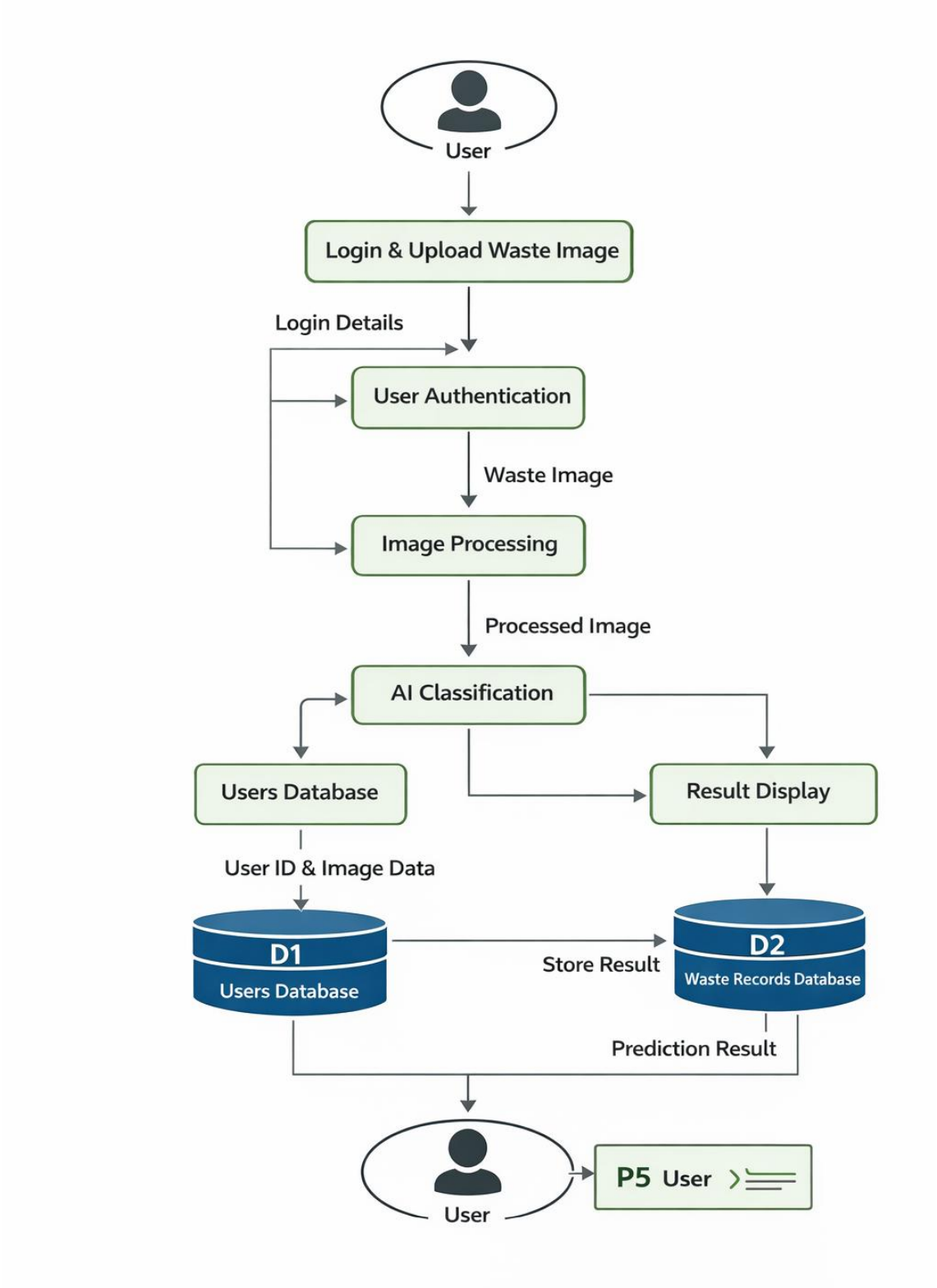

Fig A.1: BINLY – Smart AI-Based Waste Classification and Management System

B. DATABASE / TABLE STRUCTURE

Table 1: Users

Primary Key: user_id

Field Name	Data Type	Description
user_id	UUID / SERIAL	Unique user identifier
name	VARCHAR (100)	User full name
email	VARCHAR (150)	Unique email for login
password_hash	TEXT	Encrypted password
role	VARCHAR (50)	User role (Admin/User)
created_at	TIMESTAMP	Account creation time

Table 2: Waste_Records

Primary Key: record_id

Field Name	Data Type	Description
record_id	UUID / SERIAL	Unique waste record ID
user_id	UUID	Foreign key from Users table
image_url	TEXT	Uploaded waste image path
predicted_category	VARCHAR (100)	Waste classification result
confidence_score	FLOAT	AI prediction accuracy percentage
disposal_advice	TEXT	Eco-friendly disposal recommendation
uploaded_at	TIMESTAMP	Image upload date & time

Table 3: Waste_Categories

Primary Key: category_id

Field Name	Data Type	Description
category_id	SERIAL	Unique category ID
category_name	VARCHAR (100)	Type (Organic, Plastic, Metal etc.)
description	TEXT	Category explanation
recycle_type	VARCHAR (100)	Recyclable / Non - Recyclable

Table 4: Feedback

Primary Key: feedback_id

Field Name	**Data Type**	**Description**
feedback_id	**SERIAL**	**Unique feedback ID**
user_id	**UUID**	**Foreign key from Users table**
record_id	**UUID**	**Related waste prediction record**
rating	**INT**	**User rating (1–5)**
comments	**TEXT**	**User feedback text**
submitted_at	**TIMESTAMP**	**Feedback submission time**

C. SAMPLE CODING

1. App.tsx

```
import { Toaster } from "@/components/ui/toaster";

import { Toaster as Sonner } from "@/components/ui/sonner";

import { TooltipProvider } from "@/components/ui/tooltip";

import { QueryClient, QueryClientProvider } from "@tanstack/react-query";

import { BrowserRouter, Routes, Route } from "react-router-dom";

import { LanguageProvider } from "@/contexts/LanguageContext";

import Index from "./pages/Index";

import Auth from "./pages/Auth";

import LanguageSelect from "./pages/LanguageSelect";

import Dashboard from "./pages/Dashboard";

import Result from "./pages/Result";

import NotFound from "./pages/NotFound";

const queryClient = new QueryClient();

const App = () => (

  <QueryClientProvider client={queryClient}>

    <LanguageProvider>

      <TooltipProvider>

        <Toaster />

        <Sonner />

        <BrowserRouter>

          <Routes>

            <Route path="/" element={<Index />} />
```

```
            <Route path="/auth" element={<Auth />} />
            <Route path="/language" element={<LanguageSelect />} />
            <Route path="/dashboard" element={<Dashboard />} />
            <Route path="/result" element={<Result />} />
            <Route path="*" element={<NotFound />} />
          </Routes>
        </BrowserRouter>
      </TooltipProvider>
    </LanguageProvider>
  </QueryClientProvider>
);
export default App;
```

2. SUPABASE

```
import { serve } from "https://deno.land/std@0.168.0/http/server.ts";
const corsHeaders = {
  "Access-Control-Allow-Origin": "*",
  "Access-Control-Allow-Headers": "authorization, content-type",
};
serve(async (req) => {
  if (req.method === "OPTIONS") {
    return new Response(null, { headers: corsHeaders });
  }
  try {
```

```
const { imageBase64, voiceText, lang } = await req.json();
const LOVABLE_API_KEY = Deno.env.get("LOVABLE_API_KEY");
if (!LOVABLE_API_KEY) {
  throw new Error("API Key not configured");
}
const systemPrompt = `You are a waste classification AI expert.
Respond only with structured JSON containing wasteType, binColor, tips, and recommendations.`;
const userContent = imageBase64
  ? [
      { type: "text", text: "Classify this waste item." },
      { type: "image_url", image_url: { url: imageBase64 } },
    ]
  : [
      {
        type: "text",
        text: `User described waste item: ${voiceText}`,
      },
    ];
const response = await fetch(
  "https://ai.gateway.lovable.dev/v1/chat/completions",
  {
    method: "POST",
    headers: {
```

```
      Authorization: `Bearer ${LOVABLE_API_KEY}`,
      "Content-Type": "application/json",
    },
    body: JSON.stringify({
      model: "google/gemini-2.5-flash",
      messages: [
        { role: "system", content: systemPrompt },
        { role: "user", content: userContent },
      ],
    }),
  }
);
const data = await response.json();
const content = data.choices?.[0]?.message?.content;
const cleanContent = content.replace(/```json/g, "").replace(/```/g, "").trim();
const result = JSON.parse(cleanContent);
return new Response(JSON.stringify(result), {
  headers: { ...corsHeaders, "Content-Type": "application/json" },
});
} catch (error) {
return new Response(
  JSON.stringify({ error: "Waste analysis failed" }),
  { status: 500, headers: { ...corsHeaders, "Content-Type": "application/json" } }
);  } });
```

D. SAMPLE INPUT SCREENS

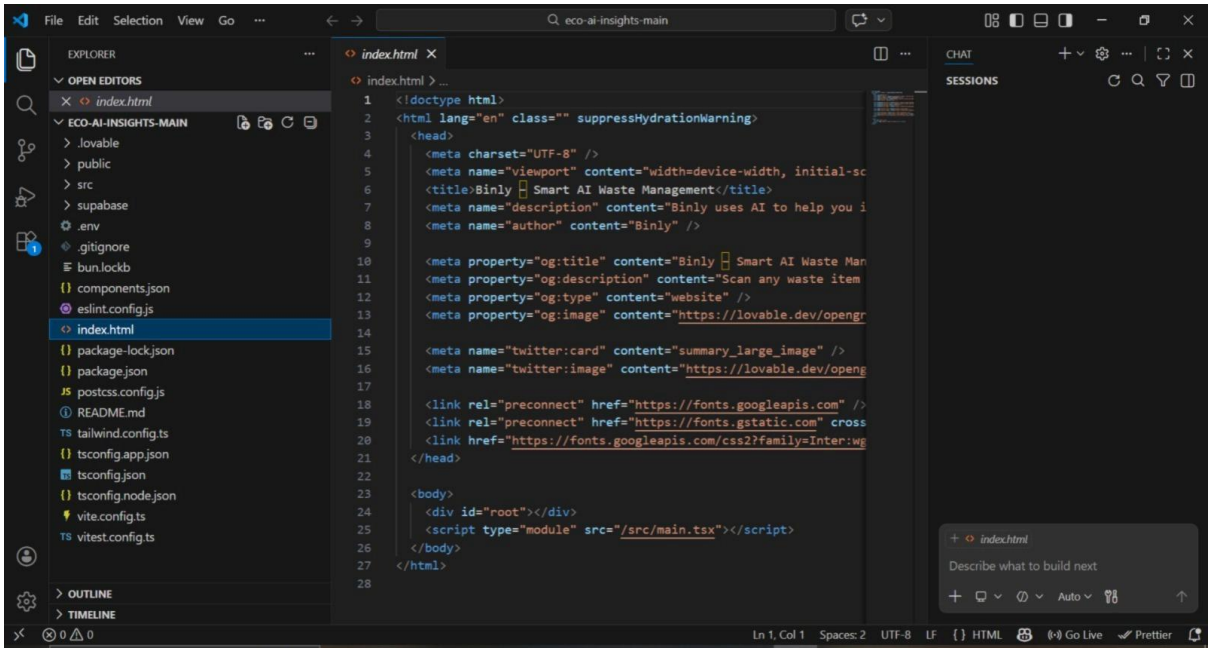

Fig D.1: Binly Terminal Screen

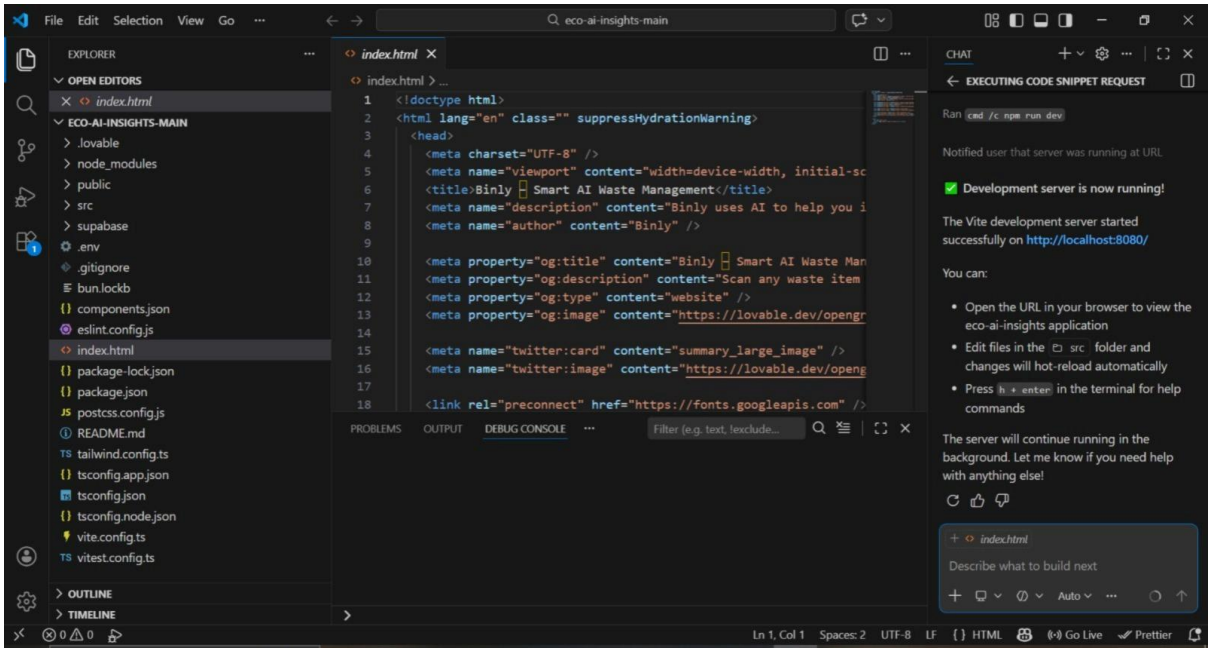

Fig D.2: Binly Frontend Terminal Screen

E. SAMPLE OUTPUT SCREENS

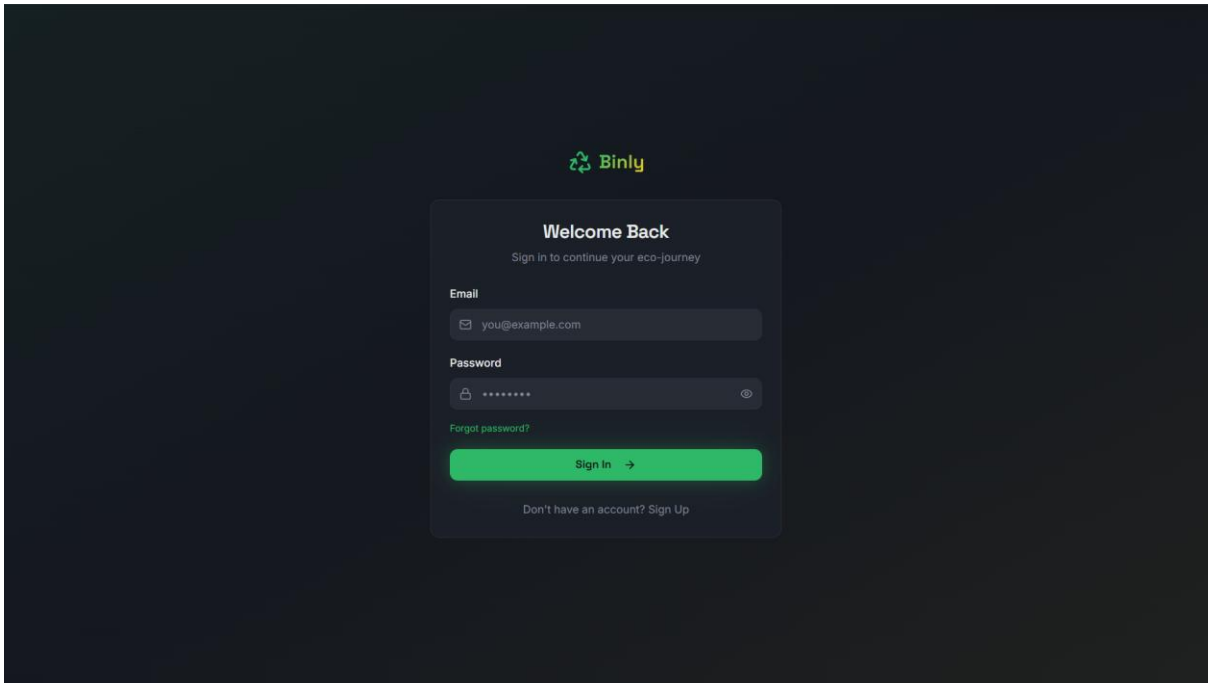

Fig 1: Binly Login Page

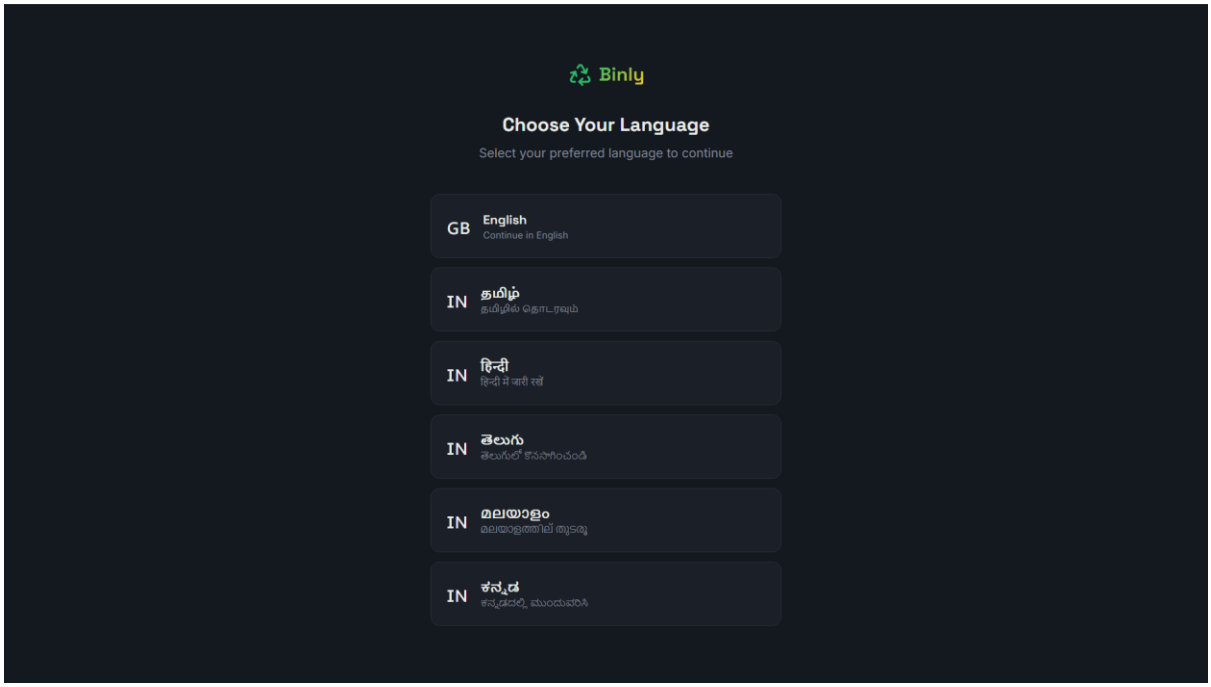

Fig 2: Language Selection Interface

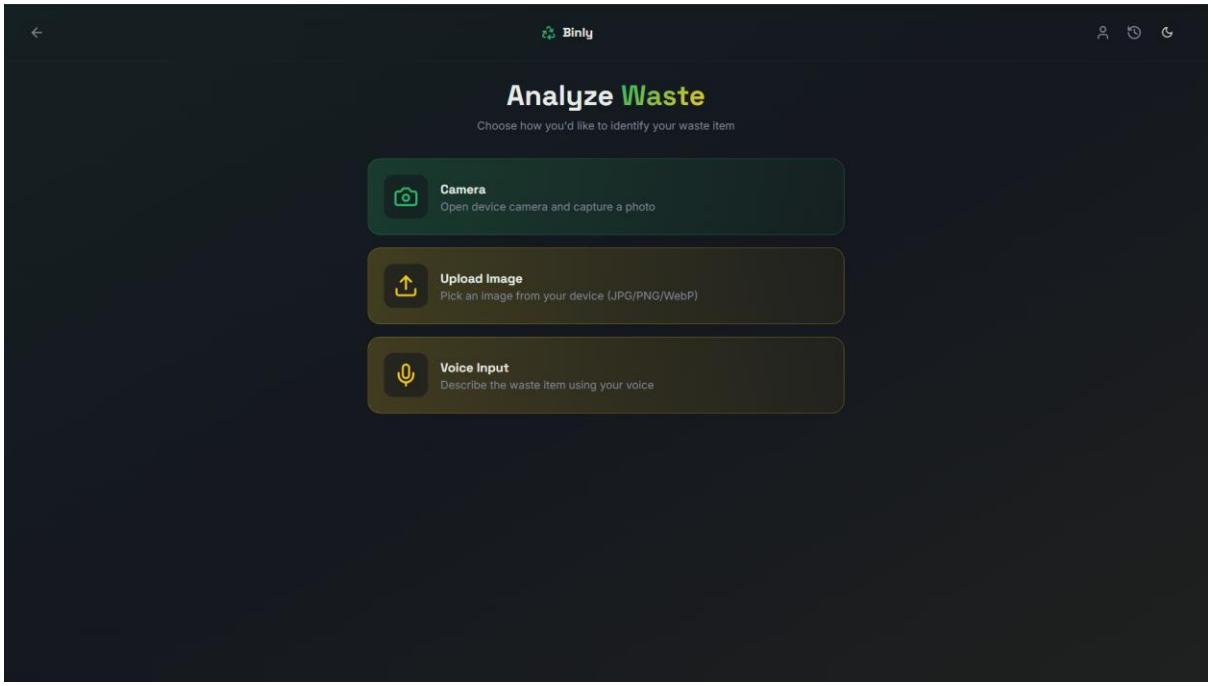

Fig 3: Waste Image Upload Page

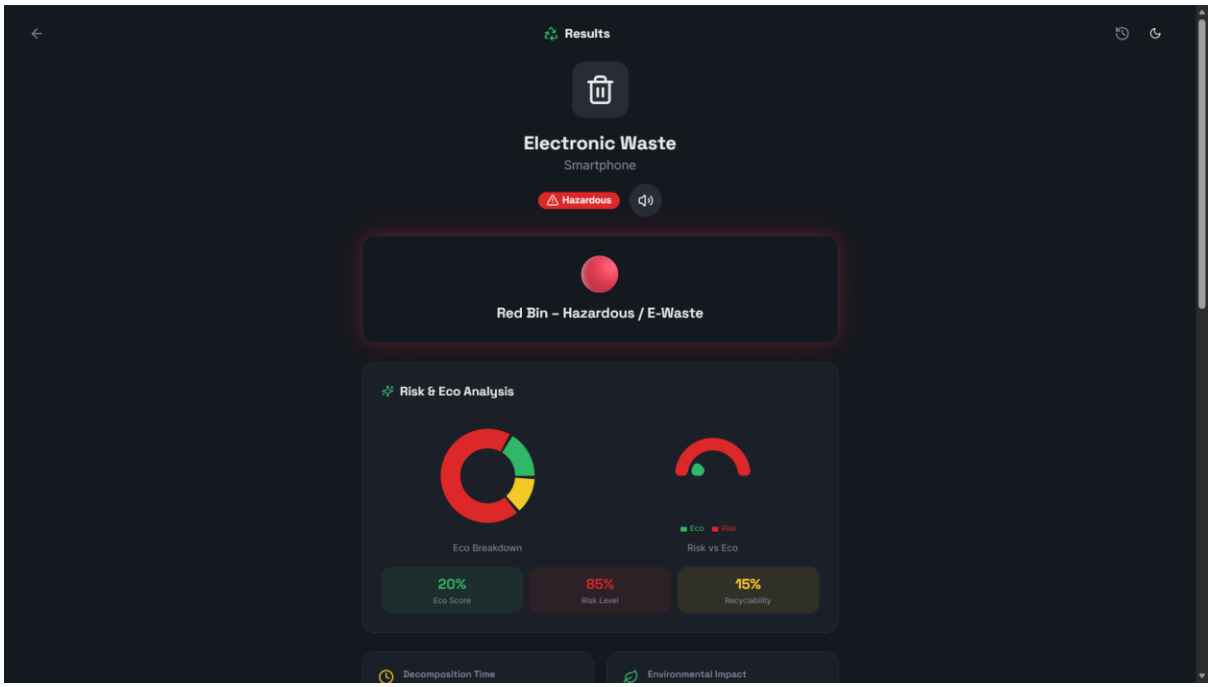

Fig 4: AI Classification Result

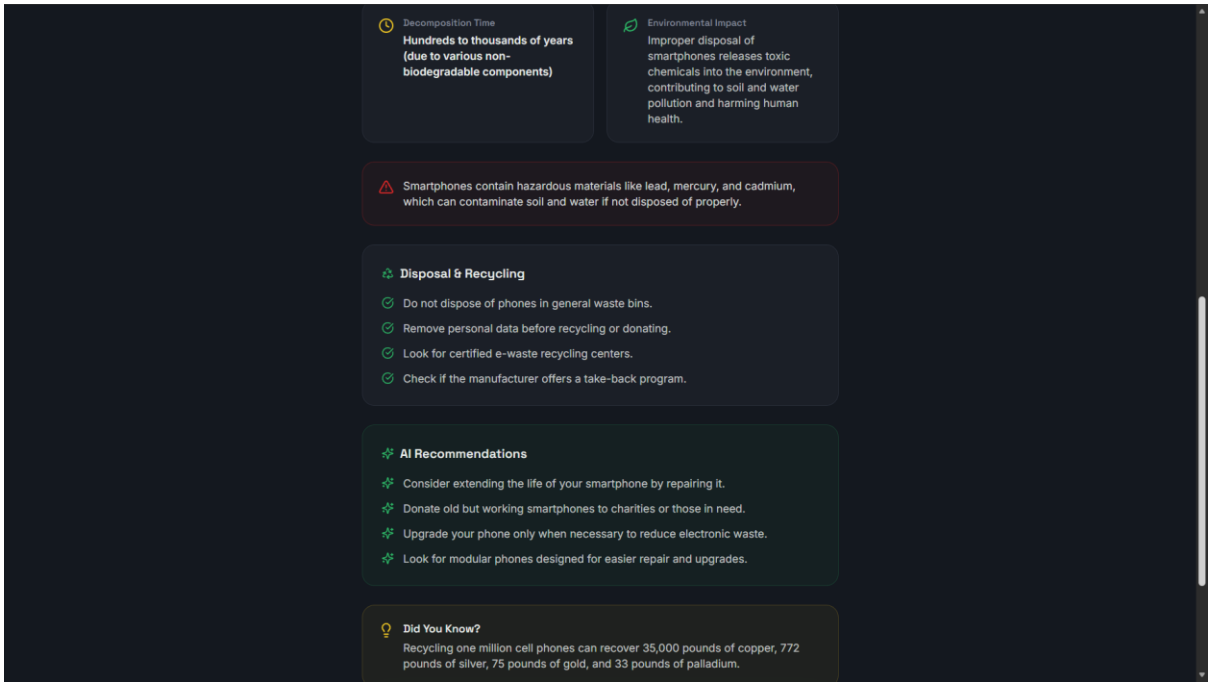

Fig 5: Waste Disposal Recommendations

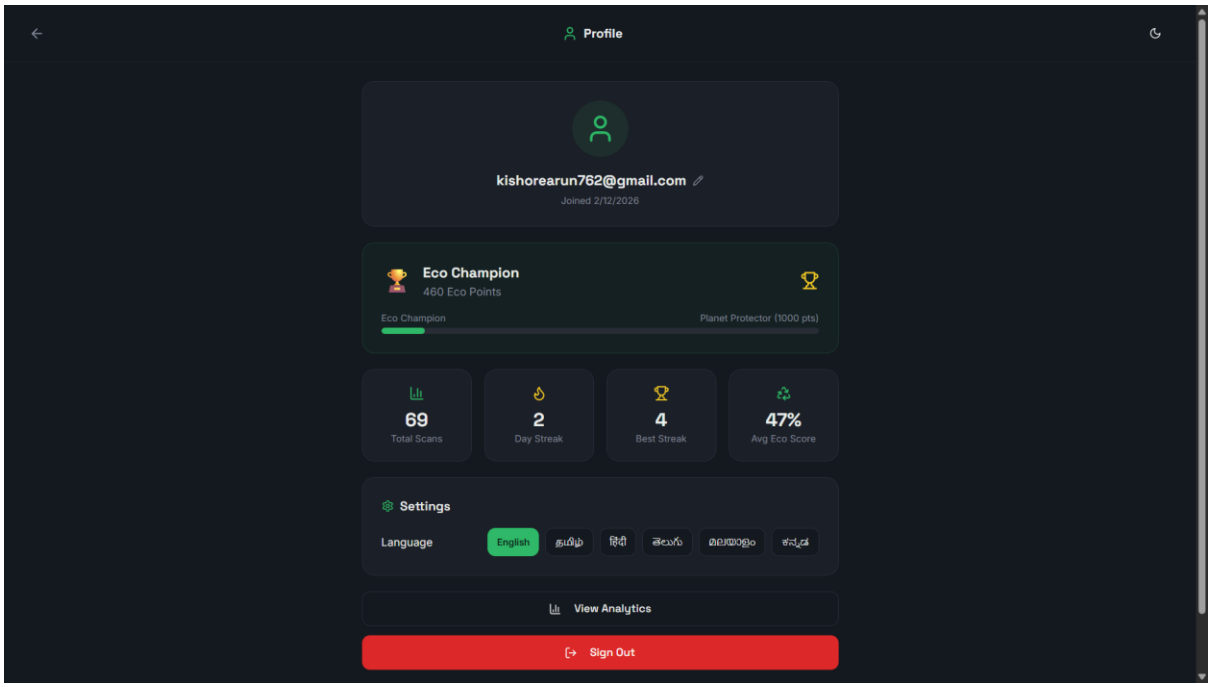

Fig: Binly User Profile

F. SYSTEM ARCHITECTURE DIAGRAM

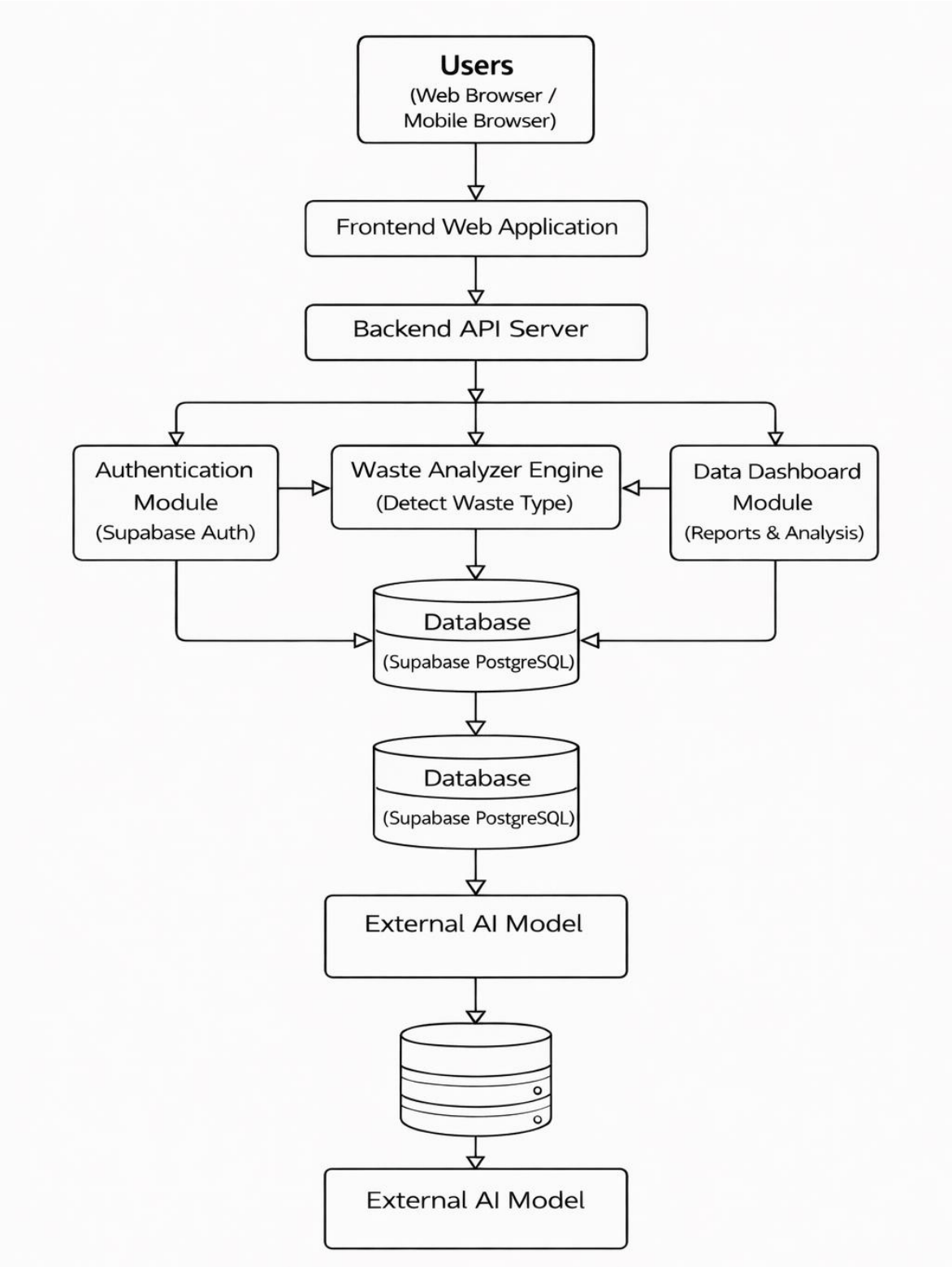

Fig: System Architecture Diagram

G. PROCESS FLOW DIAGRAM

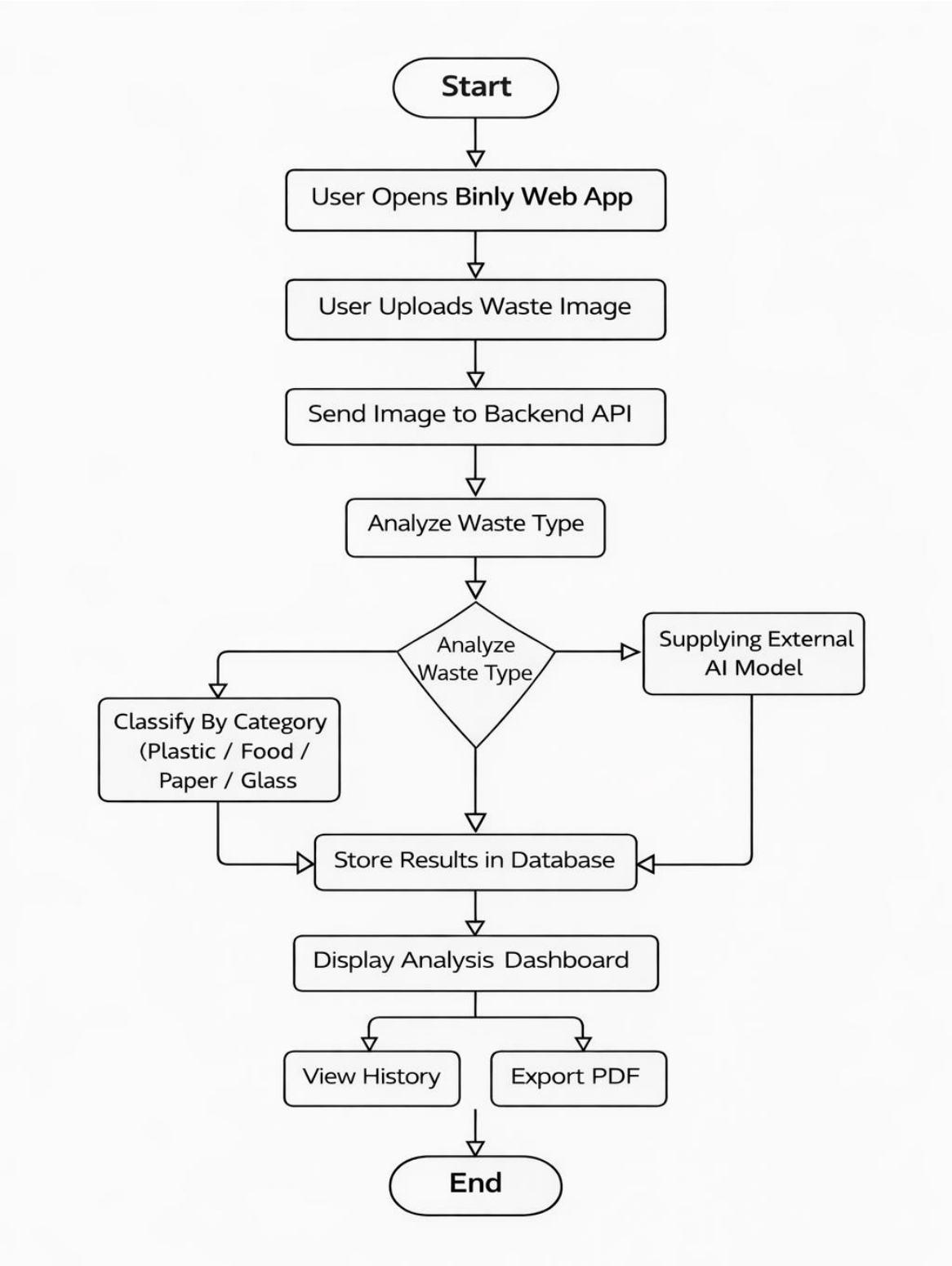

Fig: Process Flow Diagram